

Peer Review: IT Project Refinement and Ethical Integration

Reviewer: Josue Neiculeo

Classmate reviewed: Jesse Moroni Milton

Project title: Heritage Vault 2.0

Date: September 25, 2026

1. Technical Skill Assessment

Heritage Vault 2.0 is a system architecture proposal for secure genealogical record storage, and the refinement clearly builds on a strong original security and encryption foundation. The refinements themselves — a tiered visibility system (public / family-only / private), a consent flag tied to records of living individuals, continuous security monitoring with audit logging, accessibility requirements, and a "vault successor" data inheritance role — are well-scoped and each maps cleanly onto a specific problem named earlier in the document (scalability, accessibility, and privacy for living individuals).

Strengths: the tiered visibility model is a genuinely useful design pattern for this kind of system — it avoids the common all-or-nothing sharing trap that a lot of family history tools fall into. Tying the security monitoring refinement directly back to a real, named data breach (MyHeritage) rather than a generic "we added security" statement shows the refinement was driven by an actual case, not just a checklist.

Gaps: as written, this is still a proposal-level description rather than an implementation, so it's hard to evaluate some technical claims directly. A few places where more concrete detail would strengthen it: how is the tiered visibility actually enforced at the data layer (row-level permissions? separate access tables?) and how does the system prevent a "family-only"

record from leaking through a shared ancestor connection to someone outside that family group? Similarly, the scalability challenge is named clearly, but the refinement section doesn't describe a specific data model change (e.g., graph database vs. relational, indexing strategy) that would actually solve the thousands-of-connected-records problem — right now the fix for scalability isn't as clearly specified as the fixes for the other three challenges.

2. Ethical Integration Evaluation

This is the strongest part of the submission. The consent question — that a living relative's grandmother "didn't necessarily agree to have her information stored in a system she doesn't control" — is a specific, well-articulated ethical problem, not a generic privacy statement.

Tying it explicitly to the MyHeritage breach analysis from earlier in the course gives the ethical reasoning real grounding rather than treating ethics as a separate add-on section.

The data-ownership-after-death consideration is also a genuinely underexplored angle for this type of system, and the "vault successor" feature is a thoughtful, concrete response to it rather than just a stated concern with no design answer.

One angle I think could be developed further: the consent flag requires acknowledgment before a living relative's record is shared "beyond the immediate family group," but it's not clear who is empowered to grant that consent on behalf of the living relative in question, since presumably the relative themselves may not be the one using the system. That's a real edge case for a genealogy tool specifically (the person being documented is often not the person doing the documenting), and it seems worth at least naming as an open question even if a full solution is out of scope for this refinement.

3. Constructive Feedback

Technical suggestion: consider adding a short section describing the actual data model change that addresses scalability — even a sentence naming the approach (e.g., "records will

be restructured using a graph-based model to support efficient traversal of large family trees") would make that refinement as concrete as the other four.

Ethical suggestion: name who has authority to grant consent for a living relative's record when that relative isn't the one managing the vault (e.g., the closest living family member, or the vault owner with a disclosed-limitation notice). Right now the consent flag is a strong first step, but the "who consents on someone else's behalf" question is left open.

What I plan to borrow: the idea of tying a specific refinement (security monitoring) directly back to a named real-world case (MyHeritage) rather than describing it in the abstract. In my own refinement, I connected an AI-ethics case to a design decision, but Jesse's writeup does this more tightly and consistently across multiple refinements, and I want to be that specific throughout my own next revision.

4. Overall Assessment

Heritage Vault 2.0's refinement meaningfully improves on the original proposal, both technically and ethically. The four named challenges (scalability, accessibility, privacy for living individuals, and by extension data ownership) are each met with a specific, purpose-built feature rather than a generic response, and the ethical reasoning is grounded in a real case rather than abstract principle. The main opportunity for a stronger next revision is technical specificity on the scalability fix and a clearer answer to who can exercise consent on behalf of a living person who isn't the system's user.

5. Reviewer Sign-off

Reviewer signature: Josue Neiculeo

Date submitted to discussion board: September 25, 2026